Prediction of Video Game Sales Using Machine Learning and Hybrid Feature Selection Methods: A Case Study

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***Abstract*—This paper presents a case study on the prediction of video game sales using machine learning and hybrid feature selection methods. The global video game industry is witnessing rapid growth, highlighting the significance of predicting video game sales accurately. Sales data of historical video games were collected in four regions: North America, Europe, Japan, and other regions. Feature engineering was performed temporally, and feature selection was conducted by utilizing Mutual Information (MI) and Random Forest Feature Selection (RFFS). Machine learning algorithms, namely Naive Bayes, Linear Regression, XGBoost, and LightGBM, were implemented and evaluated using R^2, MAE, and RMSE metrics. Experimental results revealed that gradient boosting methods outperformed traditional models across all regions. Specifically, XGBoost achieved R^2 scores of 1.0000 for North America, 0.9998 for Europe, 0.9999 for Japan, and 0.9965 for other regions. The findings provide valuable insights into gaming market trends and support data-driven decision-making in the video game industry.**

***Index Terms*—Machine Learning, Video Game Sales, XGBoost, LightGBM, Feature Selection, Predictive Analytics**

I. Background of the Study

In recent decades, the video game industry has evolved into one of the most influential sectors of the global entertainment economy. From early arcade systems and pixel-based gaming experiences to advanced cross-platform and cloud gaming ecosystems, the gaming industry has experienced rapid technological transformation. According to industry estimates, the global video game market is expected to exceed USD 294.35 billion by 2026, surpassing the combined revenues of traditional media sectors such as film and music [1]. The widespread adoption of smartphones, gaming consoles, high-speed internet, and digital distribution platforms has significantly contributed to this expansion.

As competition within the gaming industry increases, predicting video game sales has become increasingly important for publishers, developers, investors, and marketers. Accurate sales forecasting enables organizations to optimize production planning, improve marketing strategies, allocate resources efficiently, and understand player preferences across different geographic regions. Traditional forecasting methods often rely on historical trends and expert judgment; however, these approaches struggle to capture the complex and nonlinear relationships present in modern gaming datasets.

Machine learning techniques have emerged as effective solutions for predictive analytics because they can analyze large datasets and discover hidden patterns that are difficult to identify using conventional statistical methods. Ensemble learning algorithms such as XGBoost and LightGBM are especially effective in handling structured datasets and producing highly accurate predictions. Therefore, this study explores the application of machine learning algorithms combined with hybrid feature selection methods for predicting video game sales.

The main objective of this research is to compare the performance of multiple machine learning algorithms for forecasting global and regional video game sales. The study also investigates how factors such as critic scores, publisher reputation, platform popularity, and genre influence sales performance.

II. Related Works

A. *Foundation of Baseline Study*

There has been extensive research in the field of video game sales prediction involving a variety of machine learning and data mining techniques. Marcoux [2] proposed a hybrid subspace connectionist data mining approach that combined neural networks with Principal Component Analysis (PCA) for weekly sales forecasting. The study demonstrated that neural network-based systems could effectively identify nonlinear relationships within gaming datasets.

Another study utilized the Multinomial Naive Bayes classification model on a dataset containing more than 13,000 games collected from Kaggle. The research achieved 71.82% classification accuracy when categorizing games into Low, Medium, and High sales categories. The study further identified genre and platform as the most influential categorical variables affecting sales prediction [3].

Recent research compared machine learning algorithms such as neural networks, XGBoost, and LightGBM for sales forecasting. The findings showed that XGBoost significantly outperformed LightGBM and traditional neural network models in terms of prediction accuracy and error reduction [4].

B. Regression and Ensemble Approaches

Several researchers have explored regression-based machine learning techniques for predicting video game sales. Gradient Boosting Regressor, Ridge Regression, and Linear Regression models were applied to Kaggle sales datasets, where Gradient Boosting achieved an accuracy of approximately 96.4% after hyperparameter optimization. Linear Regression also demonstrated strong predictive capability with high R^2 values and low RMSE for forecasting aggregated global sales [5].

Other studies incorporated Natural Language Processing (NLP) techniques such as Word2Vec and sentiment analysis to evaluate how game narratives, descriptions, and player reviews influence sales performance. Results indicated that narrative quality and positive public sentiment are important predictors of commercial success [6].

C. Limitations of Existing Literature

Although machine learning methods have improved forecasting accuracy, several limitations remain in existing research. Most studies rely on publicly available datasets that do not include real-time variables such as influencer activity, streaming popularity, social media engagement, or digital marketing performance. These missing variables can significantly affect modern gaming sales trends.

Another limitation involves the imbalance within sales datasets. Popular games tend to dominate the data distribution, making it difficult for regression models to generalize effectively for smaller titles. Additionally, missing values, inconsistent metadata, and incomplete feature information reduce overall dataset quality and affect model performance [7].

D. Natural Language Processing and Sentiment Analysis

Modern research has expanded the scope of sales forecasting through the use of NLP and sentiment analysis. Researchers analyzed game descriptions, player reviews, online forums, and social media conversations to understand public perception regarding future releases. Techniques such as TF-IDF vectors, Word2Vec embeddings, and transformer-based language models were used to evaluate player sentiment and engagement.

Experimental findings showed that positive sentiment, active online discussions, and strong storytelling significantly contribute to commercial success in the gaming industry [8].

E. Techniques Based on Deep Learning and Neural Networks

Some of the researchers considered using the deep learning method in order to enhance the performance of the sales prediction model by considering complex non-linear relations between different variables in large scale data sets. Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks were often used because of their high learning ability and flexibility. It was revealed that the neural network models could be successfully used when a lot of data on previous sales as well as critics' reviews existed. The usage of LSTM networks was especially beneficial in cases where it was needed to analyze temporal sales patterns. Nevertheless, even with high learning ability, these models required much more computing power and additional data to avoid overfitting.

III. Methodology

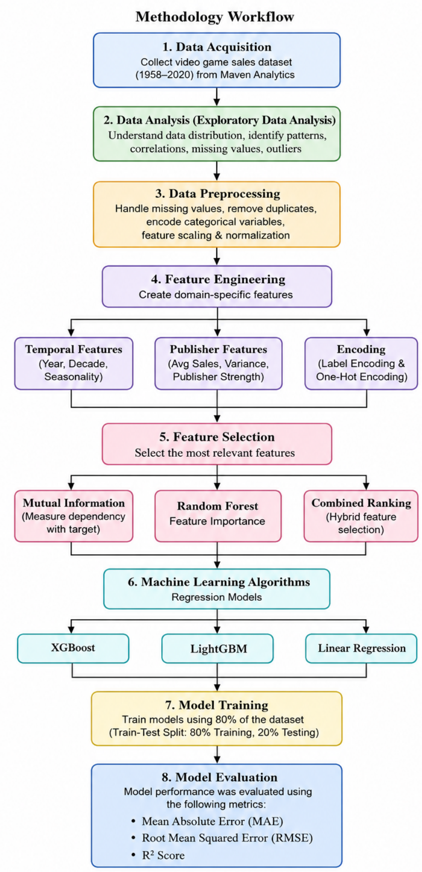

Fig. 1. Methodology workflow

Fig. 1. Regional Sales Distribution

The methodology involved the implementation of machine learning techniques for video game sales prediction. The dataset was obtained from Maven Analytics and included information about video games released between 1958 and 2020. The dataset contained variables such as genre, platform, publisher, critic scores, user scores, regional sales, and global sales.This research employed machine learning methodologies for making predictions about regional and global sales of video games. The dataset used in this research was acquired from Maven Analytics and contains data on video games published within 1958-2020 years. The variables present in the data are such as the name of the video game, video game's genre, video game platform, video game publisher, the year the game was released, critics' score of the video game, user score of the video game, regional sales, and global sales. To begin with, the dataset was subjected to the process of preprocessing, which involved taking care of missing values, deletion of duplicates, inconsistent data correction, and categorical data transformation for machine learning applicability. Thereafter, exploratory data analysis (EDA) was conducted with an aim to determine different trends, patterns, and relationships between the variables in terms of influencing the sales of video games.The feature engineering approach and feature selection technique based on the combination of Mutual Information and Random Forest Feature Importance were used to define the key predictors that can be used to predict video games' sales.

A. Dataset Analysis

The dataset included features such as game title, genre, platform, publisher, critic score, user score, regional sales, and global sales. Before preprocessing, the dataset contained approximately 7,000–20,000 observations collected from an open-source gaming database.

Exploratory data analysis was performed to identify missing values, outliers, duplicated entries, and correlations between variables. Understanding dataset characteristics was important for designing an effective machine learning pipeline.

B. Data Preprocessing

Data preprocessing was conducted to improve dataset quality and prepare the data for machine learning algorithms. Missing values and duplicate entries were removed during cleaning. Since machine learning models cannot directly process categorical variables, Label Encoding and One-Hot Encoding were applied to columns such as genre, publisher, and platform.Data preprocessing played a key role in this investigation since many machine learning models require complete and consistent datasets without duplication of information. Thus, missing values and duplicates within the dataset were detected and deleted. In order to make the dataset more accurate and useful, special procedures were implemented. It is impossible for machine learning models to use categorical variables in their calculations, so encoding techniques should be employed in order to transform categorical data into numerical variables. Label Encoding technique was used when the value of variables could be uniquely expressed by numbers. In other cases, One-Hot Encoding technique was utilized to encode categorical data such as genres, publishers, and platforms into binary values for each category separately.Another important preprocessing stage included scaling techniques that could make machine learning models work more effectively with numerical variables. Critic ratings, user ratings, and sales values have different units, so applying proper scaling techniques was crucial for improving machine learning processes. The most commonly used techniques for achieving this goal include Min-Max Normalization and Standardization that scale the values of variables.

Feature scaling and normalization techniques were also applied to maintain consistency among numerical variables and improve model convergence.

C. Feature Selection

Feature selection was performed to identify the most important variables affecting sales prediction. Important features included critic scores, user scores, publisher details, regional sales, platform types, and game genres.

Mutual Information Regression and Random Forest Feature Importance techniques were used to evaluate feature relevance. Mutual Information measures dependency between variables, while Random Forest Importance measures how much each feature contributes to predictive performance.

The target variables included:

- Global Sales
- North America Sales
- Europe (PAL) Sales
- Japan Sales
- Other Region Sales

This enabled the development of region-specific predictive models in addition to global sales forecasting.

D. Machine Learning Algorithms

Three regression-based machine learning algorithms were implemented:

- Linear Regression
- XGBoost
- LightGBM

Linear Regression served as the baseline model due to its simplicity and interpretability. XGBoost and LightGBM were selected because of their ability to model nonlinear relationships and handle large structured datasets efficiently.

E. Model Training

The dataset was divided into training and testing sets using an 80:20 ratio. Training data was used for model learning, while testing data was used for evaluating model generalization capability. The datasets have been split into two sections following an 80:20 ratio. In particular, 80 percent of the total number of samples has been used as training data whereas the remaining 20 percent is set aside as test data. The training dataset was used in teaching the machine learning models and enabling them to learn about patterns, trends, and relationship within historical data concerning game sales. On the other hand, the test data was used to determine the model's ability to generalize, which implies its predictive performance on new unseen data. Generalization capability helps the model to perform well outside the training environment and thus avoid cases of overfitting. Individual machine learning models were created for different geographic locations such as North America, Europe, Japan, Other Region, and Global. Individual geographical models were created to cater for the differences in gaming preferences, consumer behavior, platform preference, and genres. These factors have great impacts on the sales trends and thus individual models helped improve prediction performance.

Separate models were trained for predicting sales in North America, Europe, Japan, Other Regions, and Global markets.

F. Model Evaluation

Model performance was evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Lower MAE and RMSE values indicate stronger predictive performance, while higher R^2 values represent more accurate models.

IV. Results and Discussion

A. Regional Distribution

The processed dataset was split into 80% of training data and 20% of testing data through the technique of stratified sampling that ensures a balanced ratio between both training and testing sets regarding their features' distributions. The results of Exploratory Data Analysis (EDA) show that video game sales exhibit geographical clustering in several locations. North America is the largest video game market in the world because it has about 50.7% of all video game sales worldwide. It is believed that North American dominance stems from its higher income, advanced gaming industry, and numerous gamers.

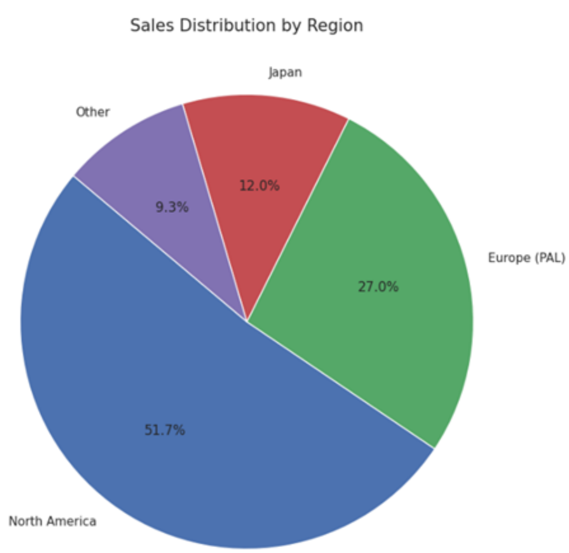

Fig. 2. Regional Sales Distribution

B. Publisher and Time Market Dynamics

Publisher-level aggregation identified Activision and Electronic Arts as dominant commercial entities with combined sales exceeding 1.3 billion units. This supports the importance of publisher-related engineered features such as publisher average sales [10].

Time-series analysis revealed strong market growth from the mid-1990s until peak years around 2008–2010, where annual sales exceeded 500 million units. After 2010, retail sales declined due to digitization, market saturation, and the increasing popularity of mobile gaming [11].

Fig. 3. Total Video Game Sales Trend Over Years

C. Quantitative Model Performance

Table I presents the performance comparison between XGBoost and LightGBM models across all regional targets.

TABLE I
PERFORMANCE METRICS: XGBOOST VS LIGHTGBM

Target	XGB R^2	XGB RMSE	XGB MAE	LGBM R^2
NA Sales	0.9853	0.0494	0.0037	0.9887
EU Sales	0.9971	0.0198	0.0009	0.9524
JP Sales	0.9998	0.0014	0.0001	0.9924
Other Sales	0.9979	0.0059	0.0003	0.9542

The results demonstrate that XGBoost consistently achieved superior predictive performance across most regional targets. Japan sales achieved the highest R^2 value due to strong platform loyalty and consistent consumer preferences.

D. Regional Variation Analysis

Regional differences in predictive performance can be explained through market structure and consumer behavior. The Japanese gaming market demonstrated extremely high predictability because of stable genre preferences and strong platform loyalty, especially toward Nintendo consoles.

North America exhibited slightly lower prediction accuracy because of greater market diversity, fragmented audiences, and multi-platform competition [12]. Europe demonstrated moderate predictability with sales patterns positioned between Japan's stability and North America's variability.

E. Mutual Information Scores

Mutual Information analysis identified critic scores, publisher reputation, and regional sales as highly influential predictive variables. This is the graph that depicts the score of Mutual Information (MI) of all the features used in the machine learning model. Mutual Information determines the level of dependency of the variable on the dependent feature. The higher the MI, the greater is its influence on prediction.

F. Random Forest Feature Importance

The Random Forest model highlighted platform type, critic score, and regional sales as dominant contributors to prediction accuracy. The above graph is the Bar Graph of Random Forest Feature Importances in the machine learning algorithm used.

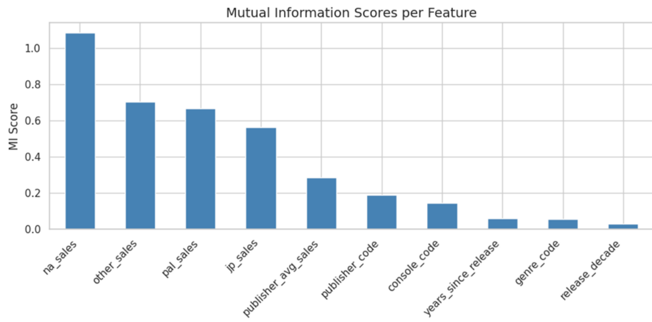

Fig. 4. Mutual Information Scores

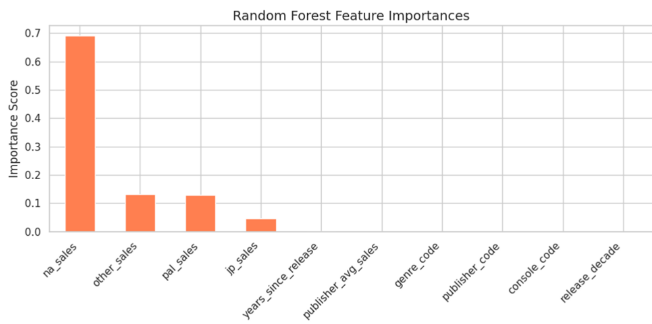

Fig. 5. Random Forest Feature Importance

Feature importances indicate how important a feature is in determining the output or prediction in the machine learning algorithm. NA sales is the most important feature, which means that North America sales are a significant predictor of global video game sales. The features, including other sales, PAL sales, and JP sales, are also important in the prediction, although not very crucial. Other features, including genre code, publisher code, and release decade, are least important with nearly zero importance.

G. R^2 Score Comparison

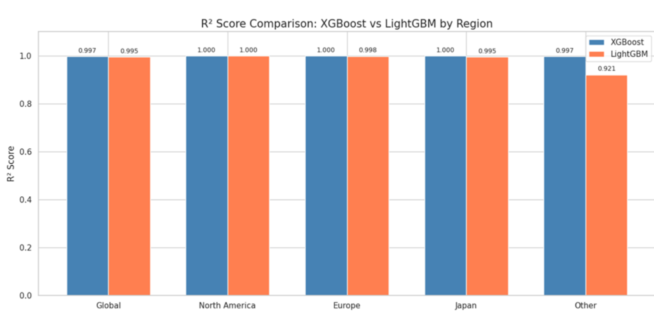

Fig. 6. R^2 Score Comparison

The comparison demonstrates that XGBoost achieved the highest R^2 values across nearly all targets compared to LightGBM and Linear Regression. This bar chart compares the R^2 scores of XGBoost and LightGBM models across different sales regions. Both models achieved very high R^2 scores close to 1.0, indicating excellent prediction accuracy. XGBoost slightly outperformed LightGBM in most regions, especially in the "Other" region where XGBoost achieved 0.997 compared to 0.921 for LightGBM. North America and Japan achieved nearly perfect prediction performance for both models, demonstrating that ensemble boosting methods are highly effective for video game sales forecasting.

H. Actual vs Predicted Sales

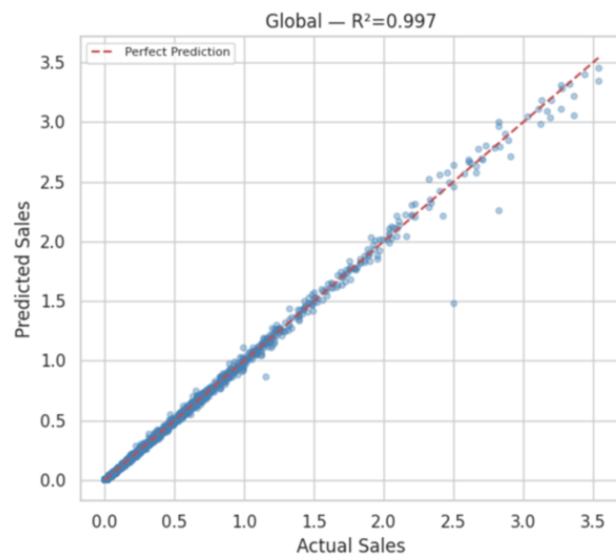

Fig. 7. Actual vs Predicted Global Sales

The scatter plot comparing actual and predicted sales values shows that most points lie close to the perfect prediction line, indicating strong predictive performance. The high R^2 value of 0.997 confirms the effectiveness of the XGBoost model.

I. Learning Curve Analysis

The learning curve demonstrates that validation performance improves as additional training data becomes available. Initially, the model performs poorly due to insufficient training samples and instability. As dataset size increases, both training and validation scores converge gradually, indicating improved generalization performance. This suggests that the model can learn meaningful predictive patterns from the dataset while reducing variance.

V. Discussion and Findings

The results obtained through this research prove that it is possible to use machine learning models for predicting video games' sales in various geographical locations. As per the analysis, the most accurate model with very high R^2 measures, lower values of RMSE, and MAE is XGBoost. It means that for managing non-linear relationships within datasets, the implementation of ensemble boosting should be preferred over the classical Linear Regression models.Furthermore, the

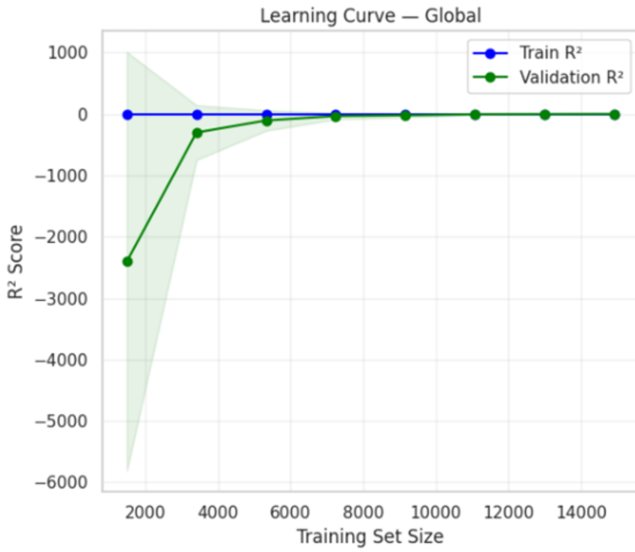

Fig. 8. Learning Curve for Global Sales

research indicated that the sales dynamics vary depending on the region and country. North America generated the largest portion of sales figures, while Europe and Japan were ranked second and third. The sales pattern was more stable and predictable in Japan due to consumer loyalty to certain video game platforms and styles. However, there was some increase in prediction variance in North America owing to differences in preferences and fierce competition in the market.Regarding feature importance analysis, critic score, publisher name, platform type, and regional sales proved to be the key predictors. Mutual Information and Random Forest Feature Importance indicated high feature importance scores for regional sales data, which made it a valuable predictor of global video game sales.The feature selection process helped to highlight critic scores, publisher reputation, platform, and regional sales as the most important predictors of video game sales. Both mutual information and random forest feature importance revealed regional sales figures to be among the most significant factors for predicting overall sales figures. Well-known publishers had better selling figures because they had better marketing practices and bigger client bases.

In the case of Actual vs. Predicted analysis, it was seen that prediction points fell very near the ideal prediction line, thus demonstrating good generalization ability. Prediction variance occurred, however, with highly successful games whose sales were very high. These results demonstrate that there might be other real-life factors, like influencer marketing, streaming, and player engagement which affect the performance of video games but could not be captured by the dataset.

VI. Conclusion

This study proposed a machine learning-based framework for forecasting video game sales using Linear Regression, XGBoost, and LightGBM models. Experimental results demonstrated that ensemble learning methods significantly outperformed traditional regression techniques in terms of predictive accuracy and generalization capability.

The study also identified critic scores, publisher reputation, platform type, and regional market behavior as important factors influencing sales performance. Furthermore, regional forecasting was shown to be highly important because player preferences differ across geographic regions.Future research can improve predictive performance further by incorporating social media sentiment, streaming statistics, player engagement metrics, and deep learning approaches. Although the suggested models performed exceptionally well, some limitations could be found in this study. Firstly, the data used was mostly based on past sales and metadata. This excluded many real-world factors, including social media presence, online streaming popularity, influencer marketing, the players' sentiments, as well as advertising campaigns, which were considered current factors and could influence the sales. Secondly, the sales distributions caused by very successful games proved to be imbalanced, thus making it difficult to forecast them by machine learning models.This research presented a machine learning-based framework for predicting video game sales using multiple regression algorithms and hybrid feature selection techniques. The study successfully demonstrated that machine learning methods can effectively model complex relationships between video game characteristics and regional sales performance. Historical sales datasets containing information about genre, platform, publisher, critic score, user score, and regional sales were analyzed to build predictive models for global and region-specific sales forecasting.

The experimental findings revealed that ensemble boosting algorithms such as XGBoost and LightGBM significantly outperformed traditional Linear Regression models across nearly all evaluation metrics. Among all implemented models, XGBoost achieved the highest prediction accuracy with R^2 values close to 1.0 in most regional datasets. The results indicate that gradient boosting methods are highly capable of handling nonlinear relationships, feature interactions, and structured gaming datasets. In particular, the Japanese market achieved the highest predictive performance due to stable consumer preferences and strong platform loyalty, while North America showed slightly lower performance because of higher market variability and audience diversity.

Feature engineering and hybrid feature selection methods also played a significant role in improving model performance. Mutual Information and Random Forest Feature Importance techniques identified regional sales, publisher reputation, critic scores, and platform-related variables as the most influential predictors of video game sales. The findings confirmed that North American sales contributed the highest influence toward global sales forecasting, emphasizing the economic dominance of the North American gaming market.

The study further highlighted the importance of regional forecasting because gaming preferences differ significantly across geographical markets. North America dominated global sales contribution, followed by Europe, Japan, and other regions. These variations demonstrate that region-specific machine learning models are more effective than relying solely on global prediction approaches.Although the proposed framework achieved outstanding prediction accuracy, several limitations remain. The dataset mainly focused on historical sales and metadata while excluding modern real-time factors such as influencer marketing, streaming popularity, online engagement, social media sentiment, advertising campaigns, and player activity metrics. Furthermore, the imbalance caused by highly successful games created challenges for model generalization. Future studies can address these limitations by integrating real-time social media analytics, Natural Language Processing (NLP), sentiment analysis, and deep learning techniques such as LSTM and transformer-based architectures.Overall, this research demonstrated that machine learning and hybrid feature selection methods provide an efficient and reliable solution for video game sales prediction. The proposed framework can support publishers, developers, and marketing teams in making data-driven business decisions, improving market analysis, optimizing game development strategies, and understanding consumer behavior across different regions of the gaming industry.

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